PRIME-CORRECTED CLOCKS, SYNTHETIC ZETA FUNCTIONS, AND SPECTRAL SPIRALS: EXACT IDENTITIES, LOCAL CERTIFICATION THEOREMS, AND UNRESOLVED BARRIERS

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Abstract. We construct a self-contained analytic framework for a prime-corrected Riemann–von Mangoldt clock and its associated synthetic zeta function. The central outputs are *proved* identities: a finite-Euler phase and residue formula, individual (rather than averaged) ordinal-cell criteria, a globally order-preserving prime flow, an exact Poisson decomposition for the corrected clock, an absolutely convergent rational-frequency expansion of its first harmonic with Möbius coefficients, and spectral moment/Rouché tests for real simple roots. For close zeros we prove a semidefinite Hankel criterion that certifies the reality of a block of zeros without simplicity, an explicit fourth-order accuracy theorem for analytically isolated pairs, and a computable two-zero certificate that takes no zero as input; an explicit quartic example shows why local accuracy cannot exclude other non-real zeros. Each theorem states its hypotheses; none asserts that the corrected clock locates every genuine zeta zero. The main text contains proofs, while an appendix records failed arguments, unverified numerical certificates and known RH-equivalent obstacles. The criterion of success is a *per-index separation threshold at every index to infinity*, never root-mean-square error.

Contents

1. Definitions and the individual objective 1
2. Synthetic zeta and the Euler-phase spiral 2
3. Analytic Abel prime correction and an ordered flow 4
4. Exact clock–prime decomposition and rational resonances 5
5. Spectral correction of close groups: separate from the synthetic argument 8
6. Clusters, real blocks and analytically isolated pairs 10
6.1. Close ordinates cannot be discarded 10
6.2. Reality of blocks without simplicity 10
6.3. Fourth-order accuracy for analytically isolated pairs 10
6.4. A computable two-zero certificate 12
6.5. Local accuracy does not exclude other non-real zeros 13
6.6. Statistical heuristics and a regularity hypothesis 13
7. What is proved and what remains an actual research problem 14
Appendix A. Arguments to defeat: failed routes, gaps, and known walls 14
Appendix B. Reported numerical cases (not certifications) 16
References 16

1. Definitions and the individual objective

Put

$$\vartheta(t) = \Im \log \Gamma\Big(\tfrac{1}{4} + \tfrac{it}{2}\Big) - \tfrac{t}{2}\log \pi, \quad N_0(t) = 1 + \frac{\vartheta(t)}{\pi}, \quad \chi(s) = \pi^{s-1/2}\frac{\Gamma((1-s)/2)}{\Gamma(s/2)} \tag{1.1}$$

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(see [13, 2]; N_0 is the smooth part of the Riemann–von Mangoldt formula [7]). The continuous logarithm is fixed by continuation along real t. On a high half-line where N_0 is increasing, the RvM clock is $\tau(\nu) = N_0^{-1}(\nu - \frac{1}{2})$; $\tau_k = \tau(k)$. The approximate clock

$$\mathcal{N}(t) = \frac{t}{2\pi}\left(\log\frac{t}{2\pi} - 1\right) + \frac{7}{8}, \qquad L(t) = \mathcal{N}'(t) = \frac{1}{2\pi}\log\frac{t}{2\pi} \tag{1.2}$$

is used *only* when stated; N_0 and $\mathcal{N}$ are not identical.

For fixed $X \geq 2$ set

$$P_X(s) = \prod_{p\leq X}(1-p^{-s})^{-1}, \qquad S_X(t) = -\frac{1}{\pi}\sum_{p\leq X}\arg(1-p^{-1/2-it}), \qquad F_X(t) = N_0(t) + S_X(t). \tag{1.3}$$

For real t, each argument is the unique principal value in $(-\pi/2, \pi/2)$. Whenever F_X is increasing, denote by κ_k the solutions $F_X(\kappa_k) = k - \frac{1}{2}$ and by m_k the solutions $F_X(m_k) = k$, with the applicable local range of indices understood.

A *synthetic system* is $0 < c_1 < c_2 < \cdots$ with $c_k \to \infty$ and $\sum c_k^{-2} < \infty$. The genuine nontrivial zeta zeros, when mentioned for evaluation, are $\rho_k = \beta_k + i\gamma_k$, indexed by *all* positive ordinates with multiplicity, without assuming $\beta_k = 1/2$.

Theorem 1.1 (Individual midpoint cell)**.** *For $k \geq 2$ put $\ell_k = c_k - c_{k-1} > 0$, $r_k = c_{k+1} - c_k > 0$, and $e_k = c_k - \gamma_k$. The statement that γ_k lies in the open Voronoi cell of c_k is* equivalent *to*

$$-\frac{r_k}{2} < e_k < \frac{\ell_k}{2}. \tag{1.4}$$

The symmetric condition $2|e_k| < \min(\ell_k, r_k)$ is sufficient, but need not be necessary. If (1.4) *holds for every k, with the first cell treated at its lower endpoint, then the γ_k are pairwise distinct; if the enumeration contains every nontrivial zero with multiplicity, this forces all nontrivial zeros onto the critical line and simple.*

Proof. The cell is $((c_{k-1}+c_k)/2, (c_k+c_{k+1})/2)$; rearranging its two inequalities yields (1.4). The open cells are disjoint. An off-line zero has a reflected partner $1-\overline{\rho}$ with the same positive ordinate, and a repeated zero gives repeated ordinates. Both contradict the strict distinctness enforced by disjoint cells. The endpoint statement follows in the same manner. □

Remark 1.2. No uniform margin is required: the ratio to the local permissible half-gap may tend to 1. Equivalently, with the directional ratio

$$R_k = \max\Big\{\frac{2e_k}{\ell_k}, -\frac{2e_k}{r_k}\Big\}, \tag{1.5}$$

condition (1.4) reads $R_k < 1$. The quantified condition is nonetheless *for every index to infinity.* Numerical root-mean-square error is irrelevant to this theorem.

2. Synthetic zeta and the Euler-phase spiral

Define

$$\Xi_c(z) = \prod_{k\geq 1}\left(1 - \frac{z^2}{c_k^2}\right), \qquad D(s) = \tfrac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2), \qquad \zeta_c(s) = \frac{\Xi_c(-i(s-1/2))}{D(s)}. \tag{2.1}$$

These are synthetic functions, not a claimed equality with Riemann's ζ.

Theorem 2.1 (Synthetic functional equation and signed zeros)**.** *The product Ξ_c is even and entire and has precisely the simple zeros $\pm c_k$. The meromorphic function ζ_c satisfies*

$$\zeta_c(s) = \chi(s)\zeta_c(1-s). \tag{2.2}$$

On the critical line

$$Z_c(t) := e^{i\vartheta(t)}\zeta_c(\tfrac{1}{2}+it) = -\frac{2\pi^{1/4}\Xi_c(t)}{(t^2+1/4)|\Gamma(1/4+it/2)|} \in \mathbb{R}, \qquad \operatorname{sgn} Z_c'(c_k) = (-1)^{k+1}. \tag{2.3}$$

Proof. On a compact set, $\sum_k |z|^2/c_k^2$ converges; hence the canonical paired product converges normally [12]. Evenness and the reality of its zeros are immediate. Under $s \mapsto 1-s$ the argument $z = -i(s-1/2)$ changes sign; direct division of $D(1-s)$ by $D(s)$ gives $\chi(s)$, proving (2.2). On $s = 1/2+it$ one has $(\frac{1}{2}+it)(-\frac{1}{2}+it) = -(t^2+\frac{1}{4})$, so

$$D(\tfrac{1}{2}+it) = -\tfrac{1}{2}(t^2+\tfrac{1}{4})\pi^{-1/4}|\Gamma(\tfrac{1}{4}+\tfrac{it}{2})|\, e^{i\vartheta(t)}, \tag{2.4}$$

which gives the closed form in (2.3). Finally

$$\Xi_c'(c_k) = -\frac{2}{c_k}\prod_{j\neq k}\Big(1-\frac{c_k^2}{c_j^2}\Big) \tag{2.5}$$

has exactly $k-1$ negative factors, hence sign $(-1)^k$; the negative prefactor in (2.3) reverses it. □

Remark 2.2 (Normalization). Since D has a simple zero at $s=1$ with $D'(1) = \frac{1}{2}$, and its only other zeros or poles are the poles at $s = -2, -4, \dots$, the function ζ_c has a single pole, at $s=1$, with

$$\operatorname{Res}_{s=1}\zeta_c = 2\,\Xi_c(\tfrac{i}{2}) = 2\prod_{k\geq 1}\Big(1+\frac{1}{4c_k^2}\Big), \tag{2.6}$$

and zeros at $s = -2, -4, \dots$. The normalized function

$$\zeta_c^\natural(s) = \frac{\zeta_c(s)}{2\,\Xi_c(i/2)} \tag{2.7}$$

has residue 1 at $s=1$, satisfies (2.2), and changes (2.3) and (2.13) only by the positive factor $(2\Xi_c(i/2))^{-1}$; no sign changes. If $c_k = \gamma_k$ for every k (which presupposes RH), the Hadamard product of ξ [13] gives $\Xi_c(z) = \xi(\frac{1}{2}+iz)/\xi(\frac{1}{2})$, so $\zeta_c = \zeta/\xi(\frac{1}{2})$ with $\xi(\frac{1}{2}) \approx 0.4971$, while $\zeta_c^\natural = \zeta$.

Set $E_X(s) = P_X(s) + \chi(s)P_X(1-s)$ and $B_X^c(s) = P_X(s)/\zeta_c(s)$. The function E_X is the finite-Euler-product model of Gonek [3], here with the plain product P_X in place of Gonek's smoothed exponential of a prime-power sum.

Theorem 2.3 (Euler-phase and meromorphic spiral)**.** *On the critical line, at points where the displayed quotients are defined,*

$$E_X(\tfrac{1}{2}+it) = 2e^{-i\vartheta(t)}|P_X(\tfrac{1}{2}+it)|\cos\big(\pi(F_X(t)-1)\big), \tag{2.8}$$

$$\frac{P_X}{E_X}(\tfrac{1}{2}+it) = \frac{1}{2} + \frac{i}{2}\tan\big(\pi(F_X(t)-1)\big), \tag{2.9}$$

$$B_X^c(\tfrac{1}{2}+it) = \frac{|P_X(\tfrac{1}{2}+it)|}{Z_c(t)}\, e^{i\pi(F_X(t)-1)}, \tag{2.10}$$

$$\frac{E_X}{\zeta_c}(\tfrac{1}{2}+it) = 2\Re B_X^c(\tfrac{1}{2}+it). \tag{2.11}$$

In particular

$$E_X(\tfrac{1}{2}+it) = 0 \iff F_X(t) \in \mathbb{Z}+\tfrac{1}{2}, \tag{2.12}$$

so on any interval where F_X is increasing the zeros of $t \mapsto E_X(\frac{1}{2}+it)$ are exactly the points κ_k, and they are simple. For $\varepsilon_k := F_X(c_k) - k + \frac{1}{2}$ the residue in the s-plane is

$$\operatorname{Res}_{s=1/2+ic_k} B_X^c(s) = \frac{|P_X(1/2+ic_k)|}{|\zeta_c'(1/2+ic_k)|}\, e^{i\pi\varepsilon_k}. \tag{2.13}$$

If F_X is strictly increasing, c_k belongs to the kth interval (m_{k-1}, m_k) if and only if

$$|\varepsilon_k| < \frac{1}{2}. \tag{2.14}$$

Proof. Write $P_X(1/2+it) = |P_X|e^{i\pi S_X(t)}$ and $\chi(1/2+it) = e^{-2i\vartheta(t)}$; since $1-s = \overline{s}$ on the line, $P_X(1/2-it) = |P_X|e^{-i\pi S_X(t)}$. Adding the two conjugate-phase terms gives (2.8), and division gives (2.9). Each factor of $P_X(1/2+it)$ is finite and nonzero, so $|P_X(1/2+it)| > 0$ and (2.12) follows from (2.8); where $F_X' > 0$ the cosine has simple zeros. Since $e^{i\vartheta}\zeta_c = Z_c \in \mathbb{R}$, the quotient gives (2.10) and (2.11). At $s_k = 1/2 + ic_k$, ζ_c has a simple zero and P_X is regular, so the residue is $P_X(s_k)/\zeta_c'(s_k)$. Differentiating $Z_c(t) = e^{i\vartheta(t)}\zeta_c(1/2+it)$ at the zero c_k, and using $F_X(c_k) - 1 = \varepsilon_k + k - \frac{3}{2}$,

$$\zeta_c'(s_k) = -i\, e^{-i\vartheta(c_k)} Z_c'(c_k), \qquad e^{i\pi(F_X(c_k)-1)} = i\,(-1)^k e^{i\pi\varepsilon_k}. \tag{2.15}$$

Hence, by Theorem 2.1,

$$\operatorname{Res}_{s=s_k} B_X^c = \frac{i\,|P_X(s_k)|\, e^{i\pi(F_X(c_k)-1)}}{Z_c'(c_k)} = \frac{(-1)^{k+1}|P_X(s_k)|\, e^{i\pi\varepsilon_k}}{Z_c'(c_k)} = \frac{|P_X(s_k)|\, e^{i\pi\varepsilon_k}}{|Z_c'(c_k)|}, \tag{2.16}$$

and $|Z_c'(c_k)| = |\zeta_c'(s_k)|$, proving (2.13). Finally (m_{k-1}, m_k) maps monotonically onto $(k-1, k)$, equivalent to (2.14). □

Remark 2.4. The curve $t \mapsto B_X^c(1/2+it)$ crosses the real axis at $F_X(t) \in \mathbb{Z}$, the imaginary axis at $F_X(t) \in \mathbb{Z} + 1/2$, and has poles at c_k. Between poles its angular speed is $\pi F_X'(t)$; crossing infinity requires a chosen projective continuation. This *synthetic spiral* records the arithmetic phase defect, not genuine zeta zero locations. By (2.12), the interval (m_{k-1}, m_k) is the Gram-type interval of the model E_X that contains its kth zero κ_k; we call it the kth *Gonek cell*. Thus (2.14) compares c_k with the kth zero of the Gonek model [3], and is different from (1.4).

3. Analytic Abel prime correction and an ordered flow

For a real exponent $a > 0$, put

$$Q_a(t) = (t/2\pi)^a, \qquad \mathcal{S}_a(t) = -\frac{1}{\pi}\sum_p \sum_{r\geq 1} \frac{e^{-p^r/Q_a(t)}}{r p^{r/2}} \sin(rt\log p), \qquad t > 0. \tag{3.1}$$

This is an *analytic model choice*; the exponent is not determined by a theorem of zero localization. The *one-step Casanya centres* are

$$c_{a,k}^{(1)} = \tau_k - \frac{\mathcal{S}_a(\tau_k)}{L(\tau_k)}. \tag{3.2}$$

Proposition 3.1 (Convergence and deterministic displacement)**.** *For every $a > 0$, the series* (3.1) *and its termwise derivatives converge locally uniformly for real $t > 0$; in particular $\mathcal{S}_a$ is real analytic on $(0, \infty)$. As $t \to \infty$,*

$$|\mathcal{S}_a(t)| \ll_a \frac{Q_a(t)^{1/2}}{\log Q_a(t)}, \qquad |c_{a,k}^{(1)} - \tau_k| \ll_a \frac{\tau_k^{a/2}}{(\log \tau_k)^2}. \tag{3.3}$$

The second statement is a bound on the synthetic correction*, not on $|c_{a,k}^{(1)} - \gamma_k|$.*

Proof. Write $Q = Q_a(t)$. Bounding each sine by 1,

$$\pi|\mathcal{S}_a(t)| \leq \sum_p \frac{e^{-p/Q}}{\sqrt{p}} + \sum_p \sum_{r\geq 2} \frac{e^{-p^r/Q}}{r\, p^{r/2}}. \tag{3.4}$$

By partial summation with $\pi(u) \ll u/\log u$ [10], the first sum is $\ll \int_2^\infty u^{-1/2} e^{-u/Q} (\log u)^{-1}\, \mathrm{d}u \ll Q^{1/4} + \sqrt{Q}/\log Q$ (split at $u = \sqrt{Q}$). The second sum is $\ll \sum_p p^{-1} e^{-p^2/Q} + \sum_p p^{-3/2} \ll \log\log Q + 1$. On a compact t-interval in $(0, \infty)$, derivatives introduce only powers of p^r, $\log p$ and $Q_a(t)^{-1}$, still

dominated by the exponential; hence local normal convergence also holds in a sufficiently narrow complex neighbourhood, proving real analyticity. For the displacement divide by $L(\tau_k) \asymp \log \tau_k$ and use $\log Q_a(\tau_k) \asymp_a \log \tau_k$. □

A construction that *proves* synthetic order, rather than assuming it, uses a flow:

$$V_a(t) = -\frac{(1-2\pi/t)^2}{L(t)}\mathcal{S}_a(t), \quad \partial_\lambda Y_a(\lambda,\nu) = V_a(Y_a(\lambda,\nu)), \quad Y_a(0,\nu) = \tau(\nu). \tag{3.5}$$

Theorem 3.2 (Global ordered prime flow)**.** *For every $0 < a \le 2$, every ν with $\tau(\nu) > 2\pi$, and each finite real λ, equation (3.5) has a unique solution in $(2\pi,\infty)$. The map $\nu \mapsto Y_a(1,\nu)$ is strictly increasing, and where $\tau'(\nu) > 0$,*

$$\partial_\nu Y_a(1,\nu) = \tau'(\nu)\exp\!\left(\int_0^1 V_a'(Y_a(v,\nu))\,\mathrm{d}v\right) > 0. \tag{3.6}$$

Thus $c_{a,k}^{\mathrm{flow}} = Y_a(1,k)$ are strictly ordered for all admissible k.

Proof. By Proposition 3.1, $\mathcal{S}_a$ is real analytic on $(0,\infty)$. The function $L(t) = (2\pi)^{-1}\log(t/2\pi)$ has a simple zero at 2π and no other zero, while $(1-2\pi/t)^2$ has a double zero there; hence V_a extends to a real-analytic function on $(0,\infty)$ with $V_a(2\pi) = 0$. In particular V_a is locally Lipschitz and 2π is an equilibrium, so by uniqueness no solution starting in $(2\pi,\infty)$ reaches 2π in finite time, forward or backward. At infinity, Proposition 3.1 gives $|V_a(t)| \ll t^{a/2}/(\log t)^2 = O(t)$ for $a \le 2$, excluding finite-time escape. Distinct orbits of a scalar autonomous equation with locally Lipschitz field cannot cross, so $\nu \mapsto Y_a(1,\nu)$ is strictly increasing because τ is. Differentiating the flow with respect to the initial parameter yields the variational equation, whose integration gives (3.6). □

Remark 3.3. This is an unconditional theorem about *synthetic order only*. It is not a localization theorem: even an ordered sequence can fail (1.4). The multiplier in (3.5) regularizes the boundary and is not uniquely imposed by arithmetic.

4. Exact clock–prime decomposition and rational resonances

For fixed X the function F_X is real analytic on the high half-line, because ϑ and each $\arg(1 - p^{-1/2-it})$ are.

Theorem 4.1 (Half-integer Poisson clock)**.** *Suppose $F_X'(t) > 0$ throughout $[T,2T]$. Let κ be its inverse at half-integer levels. As distributions on $(T,2T)$,*

$$\sum_{\kappa_k \in (T,2T)} \delta_{\kappa_k}(t) = F_X'(t)\left[1 + 2\sum_{m\ge 1}(-1)^m \cos(2\pi m F_X(t))\right]. \tag{4.1}$$

For $w \in C_c^\infty((T,2T))$, define $w_e(t) = w(|t|)$ on its two symmetric support windows and $\widehat{w}_e(u) = \int_{\mathbb{R}} w_e(t)e^{-iut}\,dt$. Then

$$\sum_k w(\kappa_k) = \int_T^{2T} w(t)N_0'(t)\,\mathrm{d}t - \frac{1}{2\pi}\sum_{p\le X}\sum_{r\ge 1}\frac{\log p}{p^{r/2}}\widehat{w}_e(r\log p) + \mathcal{R}_X[w], \tag{4.2}$$

where

$$\mathcal{R}_X[w] = 2\sum_{m\ge 1}(-1)^m \int_T^{2T} w(t)F_X'(t)\cos(2\pi m F_X(t))\,\mathrm{d}t \tag{4.3}$$

converges absolutely: with $g = w \circ F_X^{-1} \in C_c^\infty(\mathbb{R})$,

$$\int_T^{2T} w(t)F_X'(t)\cos(2\pi m F_X(t))\,\mathrm{d}t = \int_{\mathbb{R}} g(u)\cos(2\pi m u)\,\mathrm{d}u = O_{N,w}(m^{-N}) \quad \textit{for every } N. \tag{4.4}$$

Proof. The substitution $u = F_X(t)$ is a real-analytic increasing diffeomorphism of $[T, 2T]$ onto its image, so $g \in C_c^\infty$ and (4.4) follows by integration by parts in u. Poisson summation at the levels $j + \frac{1}{2}$ gives

$$\sum_k w(\kappa_k) = \sum_{j\in\mathbb{Z}} g(j+\tfrac{1}{2}) = \sum_{m\in\mathbb{Z}} (-1)^m \widehat{g}(2\pi m), \tag{4.5}$$

which, after undoing the substitution, is (4.1) tested against w. Differentiation of the finite Euler phase gives

$$S_X'(t) = -\frac{1}{\pi} \sum_{p\le X} \sum_{r\ge 1} \frac{\log p}{p^{r/2}} \cos(rt \log p), \tag{4.6}$$

an absolutely convergent series for fixed X. Splitting the $m = 0$ term as $\int w N_0' + \int w S_X'$ and using $\int_0^\infty w(t)\cos(ut)\,\mathrm{d}t = \widehat{w}_e(u)/2$ gives (4.2). □

Lemma 4.2 (Off-resonance first derivative estimate)**.** *For $m \ge 1$, real u, and $A = wF_X'$, put $\Phi_m(t) = 2\pi m F_X(t) - ut$ and $\Delta_m = \inf_{t\in\operatorname{supp} w} |\Phi_m'(t)|$. If $\Delta_m > 0$, then*

$$\left| \int_T^{2T} A(t) e^{i\Phi_m(t)}\,\mathrm{d}t \right| \le \frac{\|A'\|_1}{\Delta_m} + \frac{2\pi m \|AF_X''\|_1}{\Delta_m^2}. \tag{4.7}$$

Proof. By continuity $\Phi_m' \ne 0$ on a neighbourhood of $\operatorname{supp} A$. Integrate by parts using $e^{i\Phi} = \frac{1}{i\Phi'}\frac{d}{dt}e^{i\Phi}$; compact support removes the boundary term. Differentiate A/Φ' and use $|\Phi'| \ge \Delta_m$, $\Phi'' = 2\pi m F_X''$. □

Set $P(X) = \prod_{p\le X} p$ and

$$\mathscr{Q}_X = \{q = n/d > 0 : d \mid P(X),\ n \text{ has prime factors} \le X,\ (n,d) = 1\}. \tag{4.8}$$

For $d \mid P(X)$ define

$$c_X(d) = \prod_{p\le X,\, p\nmid d} \Big(1 - \frac{1}{p}\Big). \tag{4.9}$$

Theorem 4.3 (Exact rational-frequency decomposition with Möbius weights)**.** *For every fixed finite X and every real t,*

$$e^{-2\pi i S_X(t)} = \sum_{q=n/d\in\mathscr{Q}_X} b_X(q) q^{it}, \qquad b_X(n/d) = \frac{\mu(d) c_X(d)}{\sqrt{nd}}, \tag{4.10}$$

with absolute convergence and

$$\sum_{q\in\mathscr{Q}_X} |b_X(q)| = \prod_{p\le X} (1 + 2p^{-1/2}). \tag{4.11}$$

Moreover,

$$F_X'(t) e^{-2\pi i F_X(t)} = e^{-2i\vartheta(t)} \sum_{q\in\mathscr{Q}_X} b_X(q) \left(N_0'(t) - \frac{\log q}{2\pi} \right) q^{it}. \tag{4.12}$$

Proof. For $\alpha_p = p^{-1/2}$ and $\omega_p = p^{it}$ one has the geometric-series identity

$$\frac{1 - \alpha_p \omega_p^{-1}}{1 - \alpha_p \omega_p} = -\alpha_p \omega_p^{-1} + (1 - \alpha_p^2) \sum_{r\ge 0} \alpha_p^r \omega_p^r. \tag{4.13}$$

By $e^{-2\pi i S_X} = \overline{P_X}/P_X$ on the critical line, multiplication of (4.13) over the finitely many primes $p \le X$ yields (4.10), with negative exponent choices indexed by the squarefree denominator d and nonnegative choices indexed by n. The local absolute coefficient sum equals $\alpha_p + (1-\alpha_p^2)/(1-\alpha_p) =$

$1 + 2\alpha_p$, yielding (4.11). Absolute convergence permits termwise differentiation; differentiating $e^{-2\pi i S_X} = \sum_q b_X(q) q^{it}$ gives

$$S'_X(t) e^{-2\pi i S_X(t)} = -\frac{1}{2\pi} \sum_{q \in \mathscr{Q}_X} b_X(q) (\log q)\, q^{it}. \tag{4.14}$$

Combine this with $F'_X = N'_0 + S'_X$ and $e^{-2\pi i N_0} = e^{-2i\vartheta}$.

Equivalently, on the critical line

$$\frac{P_X(1-s)}{P_X(s)} = \Big(\sum_{d \mid P(X)} \mu(d) d^{-s} \Big) \Big(\sum_{n \ X\text{-smooth}} n^{s-1} \Big), \tag{4.15}$$

and grouping the pairs (d', n') by the reduced fraction $n'/d' = n/d$ (so $d' = de$, $n' = ne$ with $e \mid P(X)/d$) produces the weight

$$\sum_{e \mid P(X)/d} \frac{\mu(e)}{e} = \prod_{p \le X,\, p \nmid d} \Big(1 - \frac{1}{p}\Big) = c_X(d), \tag{4.16}$$

which explains the factor $c_X(d)$ in (4.10). □

Corollary 4.4 (Precise stationary-point equation)**.** *For $u \in \mathbb{R}$ and $A = wF'_X$ consider the first negative harmonic of* (4.3) *tested against e^{iut},*

$$I_u = \int_T^{2T} A(t)\, e^{i(ut - 2\pi F_X(t))}\, \mathrm{d}t, \tag{4.17}$$

whose phase is $-\Phi_1$ in the notation of Lemma 4.2. For each rational q in Theorem 4.3, the phase of the q-term obtained from (4.12) *is*

$$\Phi_{q,u}(t) = (u + \log q)t - 2\vartheta(t), \qquad \Phi'_{q,u}(t) = u + \log q - 2\vartheta'(t). \tag{4.18}$$

Thus any stationary point $t_{q,u}$ solves $2\vartheta'(t_{q,u}) = u + \log q$, and, for $t_{q,u} \to \infty$,

$$t_{q,u} = 2\pi q e^u \big(1 + O(t_{q,u}^{-2})\big). \tag{4.19}$$

Proof. Insert (4.12) into $F'_X e^{i(ut - 2\pi F_X)}$. The phase of its q-term is (4.18); the Stirling expansion $2\vartheta'(t) = \log(t/2\pi) + O(t^{-2})$ [2] proves (4.19) by inversion. □

Remark 4.5. (4.10) is a *proved exact arithmetic decomposition.* It displays squarefree Möbius denominators and X-smooth numerators without invoking genuine zeros. It does *not* yet provide a uniform stationary-phase remainder after summing over all q, especially for cutoffs $X = X(T)$. The bare single resonance $t = 2\pi e^u$ is the term $q = 1$, and by Mertens' third theorem [10] it carries the weight

$$b_X(1) = \prod_{p \le X} \Big(1 - \frac{1}{p}\Big) \sim \frac{e^{-\gamma_E}}{\log X} \qquad (X \to \infty), \tag{4.20}$$

γ_E being Euler's constant; it is the whole first harmonic only when P_X is absent.

Proposition 4.6 (Absolute monotonicity test)**.** *Put*

$$\sigma_X = \sum_{p \le X} \frac{\log p}{\sqrt{p} - 1}. \tag{4.21}$$

Then

$$|S'_X(t)| \le \sigma_X/\pi, \qquad F'_X(t) \ge N'_0(t) - \sigma_X/\pi. \tag{4.22}$$

Consequently $N'_0(t) > \sigma_X/\pi$ guarantees monotonicity at t. The prime number theorem yields $\sigma_X = (2 + o(1))\sqrt{X}$ as $X \to \infty$, so this sufficient absolute-bound argument *certifies monotonicity on a large-T window only for X of size $O((\log T)^2)$.*

Proof. With $w_p = p^{-1/2-it}$, differentiating the logarithm of each Euler factor gives

$$S'_X(t) = -\frac{1}{\pi} \sum_{p \le X} \log p \, \Re \frac{w_p}{1 - w_p}, \tag{4.23}$$

and $|\Re(w/(1-w))| \le |w|/(1-|w|)$ proves (4.22). Partial summation of $\sum_{p \le X} (\log p) p^{-1/2}$ using the prime number theorem [10] gives $2\sqrt{X} + o(\sqrt{X})$; replacing $p^{-1/2}$ by $(\sqrt{p} - 1)^{-1}$ changes the sum by $O(\sum_{p \le X} \log p / p) = O(\log X)$. □

5. Spectral correction of close groups: separate from the synthetic argument

This section *does use* the classical completed function ξ; it must not be advertised as a construction from primes and Γ alone. Put $\mathcal{F}(z) = \xi(1/2 + iz)$, a real entire function on the real axis. Its nonreal zeros occur in conjugate pairs. The following algebraic statements apply to *any* real entire function, independently of RH.

Theorem 5.1 (Two-root recovery and real Hankel criteria)**.** *Let f be real entire, $\Omega \subset \mathbb{C}$ a bounded domain invariant under conjugation with zero-free boundary $\partial\Omega$, enclosing $n \ge 1$ zeros counted with multiplicity. For $x_0 \in \mathbb{R}$ define*

$$\mathfrak{m}_j = \frac{1}{2\pi i} \oint_{\partial\Omega} (z - x_0)^j \frac{f'(z)}{f(z)} \, \mathrm{d}z, \quad \mathcal{H} = (\mathfrak{m}_{r+s})_{0 \le r,s < n}. \tag{5.1}$$

Then $\mathcal{H}$ is real symmetric,

$$\mathcal{H} \succeq 0 \quad \Longleftrightarrow \quad \text{all } n \text{ enclosed zeros are real}, \tag{5.2}$$

and

$$\mathcal{H} \succ 0 \quad \Longleftrightarrow \quad \text{all } n \text{ enclosed zeros are real and simple.} \tag{5.3}$$

For $n = 2$ the two zeros are exactly

$$z_\pm = x_0 + \frac{\mathfrak{m}_1 \pm \sqrt{2\mathfrak{m}_2 - \mathfrak{m}_1^2}}{2}. \tag{5.4}$$

Proof. This is Hermite's quadratic-form criterion [6] applied to contour moments [1, 5]. The residue theorem gives $\mathfrak{m}_j = \sum_\nu (z_\nu - x_0)^j$, sum over enclosed zeros with multiplicity; conjugation invariance makes these power sums real. For a real polynomial P of degree $< n$ with coefficient vector v,

$$v^{\mathsf{T}} \mathcal{H} v = \sum_\nu P(z_\nu - x_0)^2. \tag{5.5}$$

If all zeros are real, (5.5) is nonnegative, even when some coincide. If there is a nonreal pair $z, \bar{z}$, real Lagrange interpolation on the distinct zeros (at most n of them) produces a real P of degree $< n$ taking the values $i, -i$ at $z - x_0, \bar{z} - x_0$ and 0 at the other distinct zeros; then (5.5) equals $-2 \operatorname{mult}(z) < 0$. This proves (5.2). If all zeros are real and distinct, (5.5) vanishes only when P vanishes at n distinct points, i.e. $P = 0$; a repeated zero leaves fewer than n distinct zeros, and a nonzero P of degree $< n$ vanishing at all of them makes (5.5) zero. This proves (5.3). Finally, the quadratic formula applied to the elementary symmetric functions determined by $\mathfrak{m}_1, \mathfrak{m}_2$ yields (5.4). □

Remark 5.2. Conjugation symmetry is indispensable: a single nonreal zero in an asymmetric region gives the 1×1 matrix $(1) \succ 0$. Repeating a contour around the same set of roots only repeats its moments; turns alone create no new information.

Theorem 5.3 (Local Rouché certificate for a Taylor pair)**.** *Let f be holomorphic on a neighbourhood of $|z - x_0| \le R$, real on $\mathbb{R}$, with $x_0 \in \mathbb{R}$, and let $T_K(z) = \sum_{j=0}^{K} f^{(j)}(x_0)(z - x_0)^j / j!$ have two distinct*

real simple roots $y_- < y_+$. Suppose each closed disc $\overline{D(y_\pm, \eta)}$ is contained in $|z - x_0| \le R$ and the two discs are disjoint. Put $M_{K+1} = \sup_{|z-x_0|\le R} |f^{(K+1)}(z)|$. If

$$\frac{M_{K+1}R^{K+1}}{(K+1)!} < \min_{\substack{z\in\mathbb{C} \\ |z-y_-|=\eta \text{ or } |z-y_+|=\eta}} |T_K(z)|, \tag{5.6}$$

and each disc contains exactly one root of T_K, then each disc contains exactly one real simple *zero of f. For $K = 2$ and $\delta_0 = y_+ - y_- > 2\eta$, (5.6) is implied by*

$$\frac{M_3R^3}{6} < \frac{|f''(x_0)|}{2}\,\eta(\delta_0 - \eta). \tag{5.7}$$

Proof. Taylor's integral remainder along the straight segment in the disc gives

$$|f(z) - T_K(z)| \le \frac{M_{K+1}|z - x_0|^{K+1}}{(K+1)!} \tag{5.8}$$

(apply the one-variable integral formula to $u \mapsto f(x_0 + u(z - x_0))$). Rouché's theorem [12] on each circle gives the same zero count as T_K, namely one. As $f(\overline{z}) = \overline{f(z)}$ and each disc is symmetric about the real axis, a unique enclosed zero must be real; multiplicity one gives simplicity. For $K = 2$, $T_2(z) = \frac{f''(x_0)}{2}(z - y_-)(z - y_+)$, and on $|z - y_-| = \eta$ the other factor has modulus at least $\delta_0 - \eta$; similarly on the other circle. □

Proposition 5.4 (Near-pair accuracy of spectral Taylor truncations)**.** *Consider a family $f_h(t_* + x) = (x^2 - h^2)H_h(t_* + x)$, $t_* \in \mathbb{R}$, $h \downarrow 0$, where the analytic factors H_h are real on the real axis, nonvanishing on a common complex neighbourhood of t_*, and H_h, H_h^{-1} have derivatives of orders 0 through 5 bounded there uniformly in h. A real stationary point t_h of f_h exists near t_* and satisfies*

$$t_h = t_* + \frac{H_h'(t_*)}{2H_h(t_*)}h^2 + O(h^4). \tag{5.9}$$

If $y_\pm^{(K)}$ are the two Taylor roots near $t_* \pm h$ of order $K = 2, 3, 4$ about t_h, then

$$y_\pm^{(2)} - (t_* \pm h) = O(h^2), \quad y_\pm^{(3)} - (t_* \pm h) = O(h^3), \quad y_\pm^{(4)} - (t_* \pm h) = O(h^4). \tag{5.10}$$

Proof. Writing $t_h = t_* + x$, the critical-point equation is $G_h(x) = 0$ with

$$G_h(x) = 2xH_h(t_* + x) + (x^2 - h^2)H_h'(t_* + x). \tag{5.11}$$

By the uniform bounds there are $\eta, \rho, C > 0$ with $|H_h| \ge \eta$ and $|G_h'(x) - 2H_h(t_*)| \le C(|x| + h^2)$ for $|x| \le \rho$; hence for small h the real function G_h' keeps the sign of $H_h(t_*)$ and $|G_h'| \ge \eta$ on $|x| \le \rho$, while $|G_h(0)| = h^2|H_h'(t_*)| \le Ch^2$. Therefore G_h has a unique zero x_h in $|x| \le \rho$, with $|x_h| \le Ch^2/\eta$ (uniform version of the implicit-function theorem; no regularity in h is needed). Substituting $x_h = O(h^2)$ into $G_h(x_h) = 0$ gives $2x_hH_h(t_*) = h^2H_h'(t_*) + O(h^4)$, i.e. (5.9). The order-$K$ Taylor remainder about t_h at distance $O(h)$ is $O(h^{K+1})$, uniformly by the hypotheses, whereas $|f_h(z)| \ge 2\eta\, h\, |z - t_* \mp h|(1 + o(1))$ near the root $t_* \pm h$. Rouché on circles of radius C_1h^K around $t_* \pm h$, with C_1 large, gives a Taylor root at distance $O(h^K)$, real by conjugation symmetry. For $K = 2$ the Taylor polynomial has the two real roots $t_h \pm \sqrt{-2f_h(t_h)/f_h''(t_h)}$ for small h. □

Remark 5.5. The real stationary point method is useful for repairing *local pairs* even when the one-step Casanya centres (3.2) miss the directional cell. It does not certify their global ordinal labels, nor rule out additional nonreal zeros. The implied constants in (5.10) depend on the bounds for H_h, that is, on the distance from the pair to the other zeros; for a single pair at a fixed height the proposition gives no numerical bound, and Theorem 6.5 makes the dependence explicit. Derivative bounds must be validated on complex discs, not inferred from a mesh of sample values.

6. Clusters, real blocks and analytically isolated pairs

This section concerns the *certification* of close zeros, as opposed to their ordinal separation by a synthetic clock. It uses $\mathcal{F}(z) = \xi(\frac{1}{2} + iz)$ of Section 5 and general functions real on $\mathbb{R}$.

6.1. **Close ordinates cannot be discarded.**

Proposition 6.1 (Coincident ordinates of off-line zeros)**.** *If $\zeta(\rho) = 0$ with*

$$\rho = \beta + i\gamma, \quad 0 < \beta < 1,\ \beta \neq \tfrac{1}{2},\ \gamma > 0, \qquad \text{then} \qquad 1 - \overline{\rho} = 1 - \beta + i\gamma \tag{6.1}$$

is a distinct zero with the same positive ordinate γ.

Proof. $\xi(s) = \xi(1-s)$ and $\xi(\overline{s}) = \overline{\xi(s)}$ [13], so $\xi(1-\overline{\rho}) = \overline{\xi(\rho)} = 0$; and $1 - \overline{\rho} \neq \rho$ because $\beta \neq \frac{1}{2}$. □

Remark 6.2. Two critical zeros, however close, have distinct ordinates; two symmetric off-line zeros have ordinate separation exactly zero. Hence the closest ordinates are precisely where a violation of RH would sit, and they cannot be declared harmless. Certifying all zeros outside an exceptional set of indices, even of density zero, leaves RH undecided; and a statistical limiting law (for instance GUE) is insensitive to exceptional sets of density zero. Close pairs may therefore be exempted from *separation* by the clock, but not from *certification*: every exempted cluster needs another argument proving the reality of its zeros, for all clusters to infinity.

6.2. **Reality of blocks without simplicity.**

RH does not require simplicity. Criterion (5.2) certifies reality while leaving the internal separation of a block unresolved.

Corollary 6.3 (Block reality certificate)**.** *Let $(\Omega_k)_{k\geq 1}$ be bounded conjugation-symmetric domains with zero-free boundaries whose union contains every zero of $\mathcal{F}$, and let $\mathcal{H}_{\Omega_k}$ be the Hankel matrix* (5.1) *of $f = \mathcal{F}$ on Ω_k, of size $n_k = \mathfrak{m}_0(\Omega_k)$. If*

$$\mathcal{H}_{\Omega_k} \succeq 0 \qquad \text{for every } k, \tag{6.2}$$

then every zero of $\mathcal{F}$ is real, that is, RH holds. Simplicity is neither assumed nor concluded.

Proof. By (5.2) each block contains only real zeros of $\mathcal{F}$, and the blocks cover all zeros. A zero z of $\mathcal{F}$ corresponds to the zero $\frac{1}{2} + iz$ of ξ, which lies on the critical line exactly when $z \in \mathbb{R}$. □

Remark 6.4. Problematic indices, for instance $k, k+1$, can thus be grouped without forcing an individual separation: (6.2) for that block certifies that its zeros satisfy RH even if two ordinates are extremely close or coincide. The identity (5.5) does not, however, supply (6.2) for all blocks to infinity; that requires a new spectral inequality. Moreover the covering hypothesis of Corollary 6.3 is itself a zero-counting statement unless it is obtained by another route.

6.3. **Fourth-order accuracy for analytically isolated pairs.**

Theorem 6.5 (Fourth-order accuracy for isolated pairs)**.** *Let $t_* \in \mathbb{R}$, $h > 0$, $R > 0$, and let f be holomorphic on a neighbourhood of $\overline{D} = \{|z - t_*| \leq R\}$, real on $\mathbb{R}$, with*

$$f(z) = \big((z - t_*)^2 - h^2\big)H(z), \tag{6.3}$$

where H is holomorphic and zero-free on $\overline{D}$ and

$$m_* \leq |H(z)| \leq M_* \quad (z \in \overline{D}), \qquad B = \frac{M_*}{m_*}. \tag{6.4}$$

If

$$\frac{h}{R} \leq \frac{1}{1000B}, \tag{6.5}$$

then:

(i) *f' has exactly one zero t_b in $[t_* - h/4, t_* + h/4]$, and*

$$|t_b - t_*| \le 1.01\, B\frac{h^2}{R}; \tag{6.6}$$

(ii) *$f(t_b)f''(t_b) < 0$, and the quadratic half-separation satisfies*

$$h_b = \sqrt{-\frac{2f(t_b)}{f''(t_b)}}, \qquad |h_b - h| \le 10\, B^2\frac{h^3}{R^2}; \tag{6.7}$$

(iii) *the quantity*

$$\mathcal{K}_b = \frac{h_b^3}{|f''(t_b)|}\sup_{|z-t_b|\le 2h_b}|f^{(5)}(z)| \tag{6.8}$$

satisfies

$$\mathcal{K}_b \le 520\, B\Big(\frac{h}{R}\Big)^3; \tag{6.9}$$

(iv) *the Taylor polynomial T_4 of degree 4 of f about t_b has two real simple roots $y_\pm$ with*

$$|y_\pm - (t_* \pm h)| \le 40\, B\frac{h^4}{R^3}, \tag{6.10}$$

hence, relative to the true separation $2h$,

$$\frac{|y_\pm - (t_* \pm h)|}{2h} \le 20\, B\Big(\frac{h}{R}\Big)^3. \tag{6.11}$$

Proof. Write $x = z - t_*$. Since H is real on $\mathbb{R}$ and zero-free, it has constant sign on $[t_* - R, t_* + R]$. Cauchy's inequalities on discs of radius $R/2$ give

$$|H^{(j)}(z)| \le j!\, M_*\Big(\frac{2}{R}\Big)^j \qquad (|z - t_*| \le R/2). \tag{6.12}$$

(i) Differentiating (6.3),

$$f' = 2xH + (x^2 - h^2)H', \qquad f''(z) = 2H(z) + 4xH'(z) + (x^2 - h^2)H''(z). \tag{6.13}$$

For real $|x| \le h/4$, (6.12) gives

$$f'' = 2H(1 + \varepsilon), \qquad |\varepsilon| \le \frac{4|x||H'| + h^2|H''|}{2|H|} \le B\frac{h}{R} + 4B\frac{h^2}{R^2} < 1.01 \cdot 10^{-3}, \tag{6.14}$$

so f'' has the sign of H and $|f''| \ge 1.99\, m_*$ there. Also $|f'(t_*)| = h^2|H'(t_*)| \le 2M_*h^2/R \le 0.002\, m_*h$, while $\int_{t_*}^{t_*\pm h/4}|f''| \ge 0.49\, m_*h$; hence $f'(t_* - h/4)$ and $f'(t_* + h/4)$ have opposite signs, and f', being monotone, has exactly one zero t_b in the interval, with $|t_b - t_*| \le |f'(t_*)|/(1.99m_*) \le 1.01Bh^2/R$.

(ii) Put $\delta = t_b - t_*$, so $|\delta| \le 1.01Bh^2/R \le 1.01 \cdot 10^{-3}h$. Then $f(t_b) = (\delta^2 - h^2)H(t_b)$ has the sign of $-H$, and $f''(t_b) = 2H(t_b)(1 + \varepsilon')$ with $|\varepsilon'| \le 4B|\delta|/R + 4Bh^2/R^2$, so $f(t_b)f''(t_b) < 0$ and

$$h_b^2 = \frac{h^2 - \delta^2}{1 + \varepsilon'}. \tag{6.15}$$

Using $\delta^2/h^2 \le 1.03B^2h^2/R^2$, $4B|\delta|/R \le 4.04B^2h^2/R^2$, $4Bh^2/R^2 \le 4B^2h^2/R^2$ and $|\varepsilon'| < 10^{-5}$, we get $|h_b^2/h^2 - 1| \le 9.1B^2h^2/R^2$, and $|\sqrt{1+v} - 1| \le |v|$ gives (6.7). In particular $h_b \le 1.01h$.

(iii) Leibniz' rule applied to (6.3) gives the exact identity

$$f^{(5)}(z) = 20H^{(3)}(z) + 10(z - t_*)H^{(4)}(z) + \big((z - t_*)^2 - h^2\big)H^{(5)}(z). \tag{6.16}$$

The disc $|z - t_b| \le 2h_b$ lies in $|z - t_*| \le 3h \le R/2$. There (6.12) bounds the three terms by $960M_*/R^3$, $11520(h/R)M_*/R^3 \le 11.6M_*/R^3$ and $38400(h/R)^2M_*/R^3 \le 0.04M_*/R^3$, so

$$\sup_{|z-t_*|\le 3h}|f^{(5)}(z)| \le \frac{1000\, M_*}{R^3}. \tag{6.17}$$

With $|f''(t_b)| \geq 2m_*(1-|\varepsilon'|) \geq 1.99m_*$ and $h_b \leq 1.01h$ this gives $\mathcal{K}_b \leq (1.01)^3 \cdot 1000/1.99 \cdot B(h/R)^3 \leq 520B(h/R)^3$.

(iv) Let $r = 40Bh^4/R^3 \leq 4 \cdot 10^{-8}h$ and $D_\pm = \{|z-(t_* \pm h)| \leq r\}$, two disjoint discs. For $z \in D_\pm$ one has $|z - t_b| \leq h + |\delta| + r \leq 1.002h$, and the segment from t_b to z lies in $|w - t_*| \leq 3h$, so by (5.8) with $K = 4$ and (6.17),

$$|f(z) - T_4(z)| \leq \frac{1000M_*}{R^3} \frac{(1.002h)^5}{120} \leq 8.5\, M_* \frac{h^5}{R^3}. \tag{6.18}$$

On $\partial D_\pm$, by (6.3),

$$|f(z)| \geq m_*\, r(2h - r) \geq 1.99\, m_* rh = 79.6\, M_* \frac{h^5}{R^3}. \tag{6.19}$$

Thus $|f - T_4| < |f|$ on $\partial D_\pm$, and by Rouché T_4 has, like f, exactly one zero in each $D_\pm$. Since T_4 has real coefficients and $D_\pm$ is symmetric about $\mathbb{R}$, that zero $y_\pm$ is real and simple, and $|y_\pm - (t_* \pm h)| \leq r$, which is (6.10); dividing by $2h$ gives (6.11). □

The accuracy relative to the local separation improves as the pair compresses, provided the pair remains analytically isolated: the governing parameters are h/R and B, not the height. This is a statement of accuracy relative to the local gap, not a root-mean-square estimate.

6.4. **A computable two-zero certificate.** Theorem 6.5 explains why an identified pair is well approximated. The following test is computed from f and its derivatives alone.

Theorem 6.6 (Computable two-zero certificate)**.** *Let f be real on $\mathbb{R}$, $t_b \in \mathbb{R}$ with $f'(t_b) = 0$ and $f(t_b)f''(t_b) < 0$, and put $h_b = \sqrt{-2f(t_b)/f''(t_b)}$. Assume f is holomorphic on a neighbourhood of $|z - t_b| \leq \frac{3}{2}h_b$, let*

$$Q(z) = \frac{f''(t_b)}{2}((z - t_b)^2 - h_b^2) \tag{6.20}$$

be its quadratic Taylor polynomial at t_b, and

$$M_3 = \sup_{|z-t_b| \leq 3h_b/2} |f'''(z)|. \tag{6.21}$$

If

$$\frac{h_b M_3}{|f''(t_b)|} < \frac{84}{125}, \tag{6.22}$$

then f has exactly two zeros in $|z - t_b| < \frac{3}{2}h_b$, both real and simple, one in each disc $|z - (t_b \pm h_b)| < h_b/4$.

Proof. Since $f'(t_b) = 0$ and $f''(t_b)h_b^2/2 = -f(t_b)$, Q is the Taylor polynomial of degree 2, and (5.8) gives $|f(z) - Q(z)| \leq M_3|z - t_b|^3/6$ on the disc. On each circle $|z - (t_b \pm h_b)| = h_b/4$ one has $|z - t_b| \leq \frac{5}{4}h_b$ and

$$|Q(z)| \geq \frac{|f''(t_b)|}{2} \cdot \frac{h_b}{4} \cdot \frac{7h_b}{4} = \frac{7|f''(t_b)|h_b^2}{32}, \qquad |f(z) - Q(z)| \leq \frac{125M_3h_b^3}{384}, \tag{6.23}$$

and (6.22) is exactly the condition that the second bound is smaller than the first. On $|z - t_b| = \frac{3}{2}h_b$,

$$|Q(z)| \geq \frac{|f''(t_b)|}{2} \cdot \frac{5}{4}h_b^2 = \frac{5|f''(t_b)|h_b^2}{8}, \qquad |f(z) - Q(z)| \leq \frac{9M_3h_b^3}{16}, \tag{6.24}$$

and (6.22) implies $h_bM_3/|f''(t_b)| < 10/9$, so the second is again smaller. By Rouché, f has exactly two zeros in the large disc and one in each small disc; by conjugation symmetry the latter are real, and each is simple. □

Remark 6.7. The certificate applies to the holomorphic continuation of $Z(t)$ near the real axis, or to $\mathcal{F}$. Its inputs are $f(t_b)$, $f''(t_b)$ and a rigorous upper bound for M_3 (for instance by Cauchy's inequality from a bound on $|f|$ over a larger circle); no zero is an input. It gives an effective local separation; it has not been shown that every spectral extremum satisfies (6.22).

6.5. **Local accuracy does not exclude other non-real zeros.**

Remark 6.8 (A quartic example). The real entire function

$$f_h(z) = (z^2 - h^2)(z^2 + 4h^2), \qquad \text{zeros } -h,\ h,\ -2ih,\ 2ih, \tag{6.25}$$

is a polynomial of degree 4, so $f_h^{(5)} \equiv 0$ and every quantity of the form (6.8) vanishes: the fourth-order Taylor approximation locates the real pair exactly, yet f_h has the non-real pair $\pm 2ih$. Hence smallness of $\mathcal{K}$ at all real pairs does not imply that all zeros are real. Consistently, Theorem 6.5 does not apply to f_h: its factor $H = z^2 + 4h^2$ vanishes at distance $2h$, which forces $R < 2h$ and violates (6.5). The obstruction is not the accuracy of Taylor, but the exclusion of non-real zeros that do not take part in the pair under study.

6.6. **Statistical heuristics and a regularity hypothesis.**

Heuristic 6.9 (GUE spacing statistics)**.** Normalize the ordinate gaps by

$$s_k = (\gamma_{k+1} - \gamma_k)\,\frac{\log(\gamma_k/2\pi)}{2\pi}. \tag{6.26}$$

In the GUE (sine-kernel) model of the zeros [9, 11, 8] the nearest-neighbour spacing density satisfies

$$p(s) = \frac{\pi^2}{3}s^2 + O(s^4), \qquad \mathbb{P}(s_k < \varepsilon) \sim \frac{\pi^2}{9}\varepsilon^3 \quad (\varepsilon \to 0), \tag{6.27}$$

so among N gaps about $N\varepsilon^3$ fall below ε, and the probability that two *further* zeros lie within normalized distance ε of a given zero is

$$O(\varepsilon^8). \tag{6.28}$$

For the reported pair $k = 1\,000\,055$ (Appendix B), with γ_k replaced by $\tau_k \approx 600\,299.77$ in (6.26),

$$s_k \approx 0.0699, \tag{6.29}$$

a value fully compatible with (6.27).

Remark 6.10. By (6.27), GUE predicts arbitrarily small normalized gaps as the number of zeros grows, so no uniform positive minimal gap may be assumed. By (6.28) against (6.27), a strongly compressed pair is typically isolated from the other zeros, which is the regime of Theorem 6.5. This is an asymptotic statistical prediction, not a prohibition of triple clusters, and it does not supply the analytic bounds on H required there. Assuming GUE allows conditional asymptotic frequencies of pairs and clusters; checking it for millions or billions of zeros is numerical evidence only; proving it at the required strength is open. Even the full GUE law, as a limiting statement, tolerates exceptional sets of density zero, so it would neither give $R_k < 1$ in (1.5) for every k nor exclude finitely many multiple or off-line zeros.

Hypothesis 6.11 (Regularity of compressed pairs)**.** For every compressed pair located by a spectral extremum $t_{b,k}$ with quadratic half-separation h_k, there is an analytically determined radius $L_k > 0$ such that

$$\mathcal{K}_k = h_k^3\,\frac{\sup_{|z-t_{b,k}|\le L_k} |f^{(5)}(z)|}{|f''(t_{b,k})|} \tag{6.30}$$

satisfies

$$\mathcal{K}_k < C_k^{\mathrm{Rou}}, \tag{6.31}$$

where $C_k^{\mathrm{Rou}} > 0$ is the explicit threshold for which Theorem 5.3 with $K = 4$ certifies the two roots. The threshold need not be uniform in k.

Condition (6.31) is verifiable from spectral values and derivative bounds alone, without supplying the true zeros, and it is more specific than GUE: it asserts that the analytic structure of the function permits the certification of close pairs. Whether it can be proved from the structure of $\mathcal{F}$ is open; no proved obstruction to a suitable local formulation is known to us, and it does not follow from GUE.

Remark 6.12 (Adaptive-degree cluster scheme). Instead of a universal quadratic correction one may use a correction whose degree adapts to the cluster,

$$\text{Casanya} \longrightarrow \text{cluster detection} \longrightarrow \text{Taylor of degree } 2n \longrightarrow \text{Rouché certification.} \tag{6.32}$$

By Heuristic 6.9, highly compressed clusters of many zeros are progressively rarer, but the analytic procedure must allow every size n: exceptions cannot be excluded for being statistically improbable. The provable question becomes an inequality between the spectral derivatives and the geometry of each cluster. If it held for every cluster with adaptive n, it would give a universal localization procedure for the real zeros found; the absence of further non-real zeros would still need a separate argument (Remark 6.8), but it would not have been assumed in certifying each cluster.

Remark 6.13 (Status of the cluster statements).

Statement	Status
Ignoring a proportion of zeros without certifying them	Insufficient for RH
Ignoring their internal separation while certifying that all are critical	Sufficient for RH (Corollary 6.3)
Relative quartic Taylor error $O((h/R)^3)$ for isolated pairs	Proved (Theorem 6.5)
Local certification by (6.22)	Proved (Theorem 6.6)
GUE guarantees regularity of all pairs	Does not follow from GUE
Small $\mathcal{K}_k$ excludes other non-critical zeros	False (Remark 6.8)

The distinction to keep is *not separating* versus *not certifying*: for RH the first is allowed, the second is not.

7. What is proved and what remains an actual research problem

Theorem 2.3 and Theorem 4.3 give a new *object of investigation*: a fully arithmetic phase spiral whose first Fourier harmonic has exact rational frequencies $q = n/d$ and Möbius weights. Theorem 3.2 proves order preservation of a prime-driven synthetic transport, while Theorem 5.3, Theorem 6.5 and Theorem 6.6 certify individual spectral pairs under computable derivative inequalities, and Corollary 6.3 certifies reality of blocks without simplicity. None of these results asserts the universal cell inequalities (1.4).

The next nontrivial analytical estimate is *not* a reformulation of a zero count: it is a uniform-in-the-active-prime-range asymptotic for the *arithmetic sum of rational resonances* in (4.12), with controlled cancellation rather than the absolute mass (4.11). Such a theorem would be meaningful even independently of RH. Whether it ultimately produces a universal zero-separation law remains open. On the spectral side, the corresponding target is Hypothesis 6.11, or its adaptive-degree version in Remark 6.12, together with a covering argument that does not reduce to the classical complete count. By contrast, demanding the full explicit formula for an entire synthetic spectrum at once merely repackages the familiar RH-level spectral identification.

Appendix A. Arguments to defeat: failed routes, gaps, and known walls

The following is part of the mathematical record, not a collection of claimed theorems.

(1) **RMS optimization is the wrong objective.** An estimate may have larger RMS and still satisfy every individual cell. Earlier experiments using $a = 0.5, 0.75, 1$ for Abel smoothing showed local synthetic-gap contractions and some inversions. Reports based on $2|e_k|/\min(\ell_k, r_k)$

tested a *stronger symmetric* threshold, not the final directional inequality (1.4). Such failure counts must not be silently relabelled as directional failures.

(2) **The exponent 3/4 is not proved optimal.** The value $a = 3/4$ used in the experiments is not derived from any optimality principle. The auxiliary cost $J(a) = \log(1/a) + a^2$ considered in earlier tuning has its unique minimum at $a = 1/\sqrt{2} \approx 0.707$, not at 3/4, and weighting its two terms equally was arbitrary; no theorem links either value to an individual bound on $|c_k - \gamma_k|$.

(3) **An Abel-series displacement bound is not a true-zero error bound.** The estimate $|c_{a,k}^{(1)} - \tau_k| \ll \tau_k^{a/2}/(\log \tau_k)^2$ of (3.3) neither proves nor disproves a bound on $|c_{a,k}^{(1)} - \gamma_k|$.

(4) **Guaranteed synthetic monotonicity does not imply localization.** The flow theorem removes synthetic inversions but does not prevent wrong ordinal assignment. Retrospective tests showed residual failures near compressed pairs, including the neighbourhood of index 1 000 055; no all-k conclusion follows.

(5) **Newton and paired Taylor steps are not globally certified.** Independent Newton iterations may converge twice to the same zero. Pairwise Taylor approximations have excellent local numerical examples, but Rouché requires rigorously bounded coefficients and complex-disc derivatives. Previously quoted fifth-derivative bounds near index 1 000 055 are *not a finished interval certificate*.

(6) **A norm of logarithmic-derivative discrepancy can be overly strong.** If $f = e^{Az^2} g$ with $A \in \mathbb{R}$, then f and g have exactly the same zeros, while $(\log f)' - (\log g)' = 2Az$ can have arbitrarily large absolute contour integral. Oriented polynomial moments of the difference all vanish. Thus a large absolute norm does not alone rule out exact spectral transport.

(7) **Hankel positivity is conditional.** It is equivalent to reality and simplicity of the enclosed zeros (semidefiniteness to reality alone), but no universal positivity bound for the completed zeta function has been established. The analogous absolute transport bound deteriorates when synthetic Vandermonde matrices become ill-conditioned in close clusters.

(8) **The "even defect" route is a known wall.** For a conjugation-symmetric domain, the total zero count minus the number of certified real-axis sign changes is a nonnegative even integer. Demanding that it vanish for an exhaustive covering is a classical RH-plus-simplicity-level zero count [14], not an independent new inequality.

(9) **The full synthetic explicit formula is another known wall.** Demanding equality of the prime trace for *all* admissible test functions forces spectral identification, rather than offering a readily provable intermediate estimate. Such tests require carefully specified Fourier support and growth; a function cannot be both compactly supported in t and have compactly supported Fourier transform unless it vanishes.

(10) **The single lattice resonance was an erroneous simplification.** With the Euler phase, $F_X'' = N_0'' + S_X''$, and S_X'' is in general not small compared with $N_0'' \asymp 1/t$. Stationary points occur in an arithmetic family $t \sim 2\pi q e^u$, not solely at $2\pi e^u$. The exact repaired statement is Theorem 4.3; a summable stationary-phase error for growing X has *not* been proved.

(11) **A sharp growing cutoff is not a smooth clock.** Choosing $X = X(t)$ by suddenly admitting a new prime can make $S_{X(t)}$ jump. Proposition 4.6 yields merely a sufficient, not a necessary, monotonicity range. A monotone regularization may destroy the exact coefficient identity (4.10) unless its effect is tracked.

(12) **A full Euler product is not an entire model.** P_X has poles, χ is meromorphic, and $E_X = P_X + \chi P_X(1 - s)$ inherits meromorphic singularities in general. Replacing P_X by an exponential of a finite prime-power sum, as in [3, 4], removes its own poles, but does not automatically remove the poles of χ.

(13) **The Möbius resonance sum is not yet a prime selector.** The denominator coefficients $b_X(1/d) = \mu(d)c_X(d)/\sqrt{d}$ of (4.10) constitute a finite-sieve structure. Showing that its combined resonances produce $\Lambda(n)$ with a useful uniform error is unproved. The informal claim that a universal cutoff $X(T) \geq \sqrt{T/2\pi}$ is *necessary* is not established by the sieve heuristic.

(14) **A local spectral certificate is not RH.** A collection of rigorously certified real pairs does not exclude unobserved nonreal zeros (Remark 6.8). Proving exhaustive global coverage by ordinary argument counting [14] returns to the well-known obstruction. A genuinely new analytic inequality is required for any claim beyond local localization.

Appendix B. Reported numerical cases (not certifications)

The following figures are *retrospective observations* archived in the project; no new fitting or independent interval verification was performed here. They cannot establish statements for all indices. Each entry is the reported maximum, over the two indices of the pair, of the absolute deviation in t of the order-2 (resp. order-3) Taylor root of Proposition 5.4 from the archived ordinate.

Pair of indices	Maximum quadratic error	Maximum cubic error
889–890	0.0058411	0.0021378
1 000 055–1 000 056	0.0001344	0.0000122
2 000 992–2 000 993	0.0053030	0.0020388
2 001 043–2 001 044	0.0013431	0.0009932

The near pair 1 000 055–1 000 056 had a reported true separation of 0.038304245 (normalized gap (6.29)); the prime flow with $a = 0.75$ missed the stronger symmetric cell threshold at its first index (reported ratio $2|e_k|/\min(\ell_k, r_k) = 1.058636$), whereas an independent spectral Newton step reduced the reported ratio below 1. The fourth-order Taylor predictions were reported to have errors of order 10^{-7}, but the accompanying fifth-derivative bounds were *not certified by interval arithmetic*. Numerical corrections of thirteen previously observed flow failures by mixed Newton/pair rules were local-window tests, not a verified complete infinite sequence. These experiments motivate Theorem 5.3; they do not prove it applies to those particular numerical inputs without validated coefficient bounds. A rigorous certification of such a pair is obtained either by interval evaluation of Z at three points (two sign changes) combined with Turing's method on the surrounding window [14], or by Theorem 6.6 with a rigorous bound for M_3.

Data availability. The archived material referred to in Appendices A and B is in the public repository `https://github.com/vcasana1/LawCasanya` (branch `main`), including `https://github.com/vcasana1/LawCasanya/tree/main/work/display`. No new numerical benchmark was performed for this article.

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